

Sprint Planning and Story Point Breakdown

HDI Prediction System project example



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This document summarizes sprint planning and story point estimation for the **HDI Prediction System** project. It provides a clear breakdown of work into sprints, epics, and user stories to support predictable delivery and transparent progress tracking.

1. Introduction

A **Sprint** is a fixed period (time-box) during which a team works to complete a defined set of tasks.

Sprints help teams plan work, track progress, and deliver outcomes in manageable, repeatable cycles—supporting frequent feedback and continuous improvement.

2. Key Agile Terms

The following terms are used throughout the sprint plan for the HDI Prediction System project:

Term	Meaning (in this project)
Epic	A large piece of work that is too big to complete in one sprint, so it is broken into smaller stories delivered across multiple sprints.
Story	A small, implementable task that contributes to completing an epic.
Story Point	A number representing the effort required to complete a story (considering complexity, risk, and time). Story points are often estimated using the Fibonacci series (e.g., 1, 2, 3, 5, 8, ...).

3. Story Point Scale

The project uses the following simplified story point scale for consistent estimation:

Task Difficulty	Story Points
Very Easy	1
Normal	2
Moderate	3
Difficult	5

Note — Story points reflect relative effort rather than exact hours, helping teams estimate work consistently across sprints.

4. Sprint Breakdown for the HDI Prediction System Project

Sprint 1

Sprint 1 focuses on foundational data work: collecting relevant HDI-related indicators and preparing the dataset for modeling and analysis.

Epic	Story (ID)	Story Points
Data Collection (Epic 1)	Gathering Data (USN1)	2
Data Collection (Epic 1)	Loading Data (USN2)	1
Data Preparation (Epic 2)	Handling Missing Values (USN3)	3
Data Preparation (Epic 2)	Creating Fields (USN4)	3
Data Preparation (Epic 2)	Handling Inconsistency in Data (USN5)	3
Total Story Points in Sprint 1		12

Sprint 2

Sprint 2 focuses on communicating insights through visuals and delivering a usable dashboard experience for users.

Epic	Story (ID)	Story Points
Data Visualization (Epic 3)	Bar Chart (USN6)	2
Data Visualization (Epic 3)	Pie Chart (USN7)	2
Data Visualization (Epic 3)	Line Chart (USN8)	2
Data Visualization (Epic 3)	Map (USN9)	4
Dashboard (Epic 4)	Developing Dashboard (USN10)	5
Story	Developing Story (USN11)	5
Total Story Points in Sprint 2		20

5. Velocity Calculation

Velocity is calculated as the average number of story points completed per sprint.

Item	Value
Total Story Points	$12 + 20 = 32$
Number of Sprints	2
Velocity	$32 / 2 = 16$ story points per sprint

Result — The team's velocity is **16 story points per sprint**.

Velocity helps estimate how much work the team can complete in future sprints and improves planning accuracy as more sprint data becomes available.

6. Conclusion

The sprint, epic, story, and story point breakdown supports structured delivery for the HDI Prediction System project. By splitting work into time-boxed sprints and estimating effort using story points, the team can plan more accurately, monitor progress consistently, and deliver incremental value with each sprint cycle.